Worksheet 6

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installed packages

```
library(ISLR2)
library(survival)
library(survminer)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: ggpubr
```

```
##
## Attaching package: 'survminer'
```

```
## The following object is masked from 'package:survival':
##
##     myeloma
```

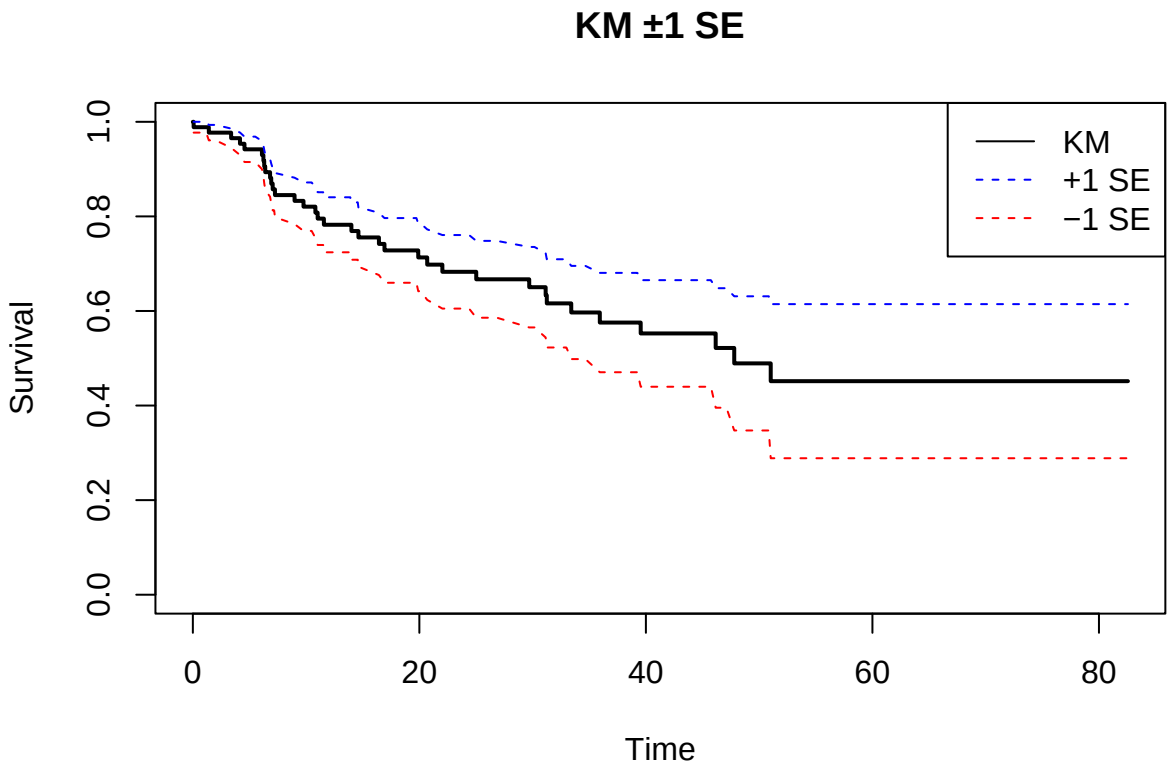
running libraries....

```
## 'data.frame':    88 obs. of  8 variables:
##  $ sex      : Factor w/ 2 levels "Female","Male": 1 2 1 1 2 1 2 2 1 2 ...
##  $ diagnosis: Factor w/ 4 levels "Meningioma","LG glioma",..: 1 3 1 2 3 1 1 2 1 3 ...
##  $ loc      : Factor w/ 2 levels "Infratentorial",..: 1 2 1 2 2 2 2 2 2 2 ...
##  $ ki       : int  90 90 70 80 90 80 80 80 70 100 ...
##  $ gtv      : num  6.11 19.35 7.95 7.61 5.06 ...
##  $ stereo   : Factor w/ 2 levels "SRS","SRT": 1 2 1 2 2 1 2 2 2 2 ...
##  $ status   : int  0 1 0 1 1 0 0 0 0 0 ...
##  $ time     : num  57.64 8.98 26.46 47.8 6.3 ...
```

```
##      sex  diagnosis           loc ki   gtv stereo status  time
## 1 Female Meningioma Infratentorial 90  6.11    SRS      0 57.64
## 2   Male  HG glioma Supratentorial 90 19.35    SRT      1  8.98
## 3 Female Meningioma Infratentorial 70  7.95    SRS      0 26.46
## 4 Female  LG glioma Supratentorial 80  7.61    SRT      1 47.80
## 5   Male  HG glioma Supratentorial 90  5.06    SRT      1  6.30
## 6 Female Meningioma Supratentorial 80  4.82    SRS      0 52.75
```

(a) Kaplan-Meier Survival Curve with ±1 SE Bands

```r
km <- survfit(Surv(time, status) ~ 1, data = BrainCancer)
plot(km, conf.int = FALSE, col = "black", lwd = 2, xlab = "Time", ylab = "Survival", main = "KM ±1 SE")
lines(km$time, pmin(km$surv + km$std.err, 1), col = "blue", lty = 2)
lines(km$time, pmax(km$surv - km$std.err, 0), col = "red", lty = 2)
legend("topright", c("KM", "+1 SE", "-1 SE"), col = c("black", "blue", "red"), lty = c(1,2,2))
```




```r
#default 95% confidence interval
```

(b) Bootstrap Estimation of SE

```r
set.seed(123)
n <- 88
B <- 200
time_points <- unique(sort(BrainCancer$time))
surv_matrix <- matrix(NA, nrow = B, ncol = length(time_points))

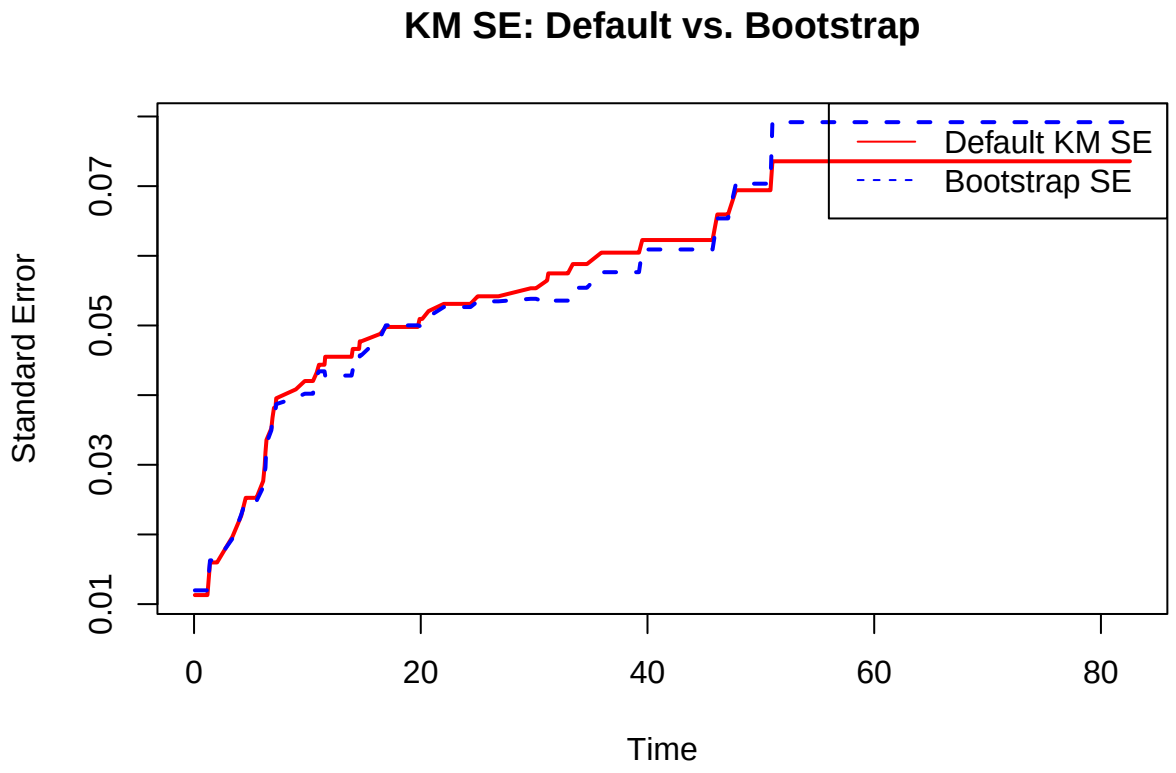
for (b in 1:B) {
  index <- sample(1:n, n, replace = TRUE)
  data1 <- BrainCancer[index, ]
  km_boot <- survfit(Surv(time, status) ~ 1, data = data1)
  surv_b <- summary(km_boot, times = time_points, extend = TRUE)$surv
  surv_matrix[b, ] <- surv_b
}

bs_se <- apply(surv_matrix, 2, sd) #std errors for bootstrapped model
```

comparing to (a)

```
#Compare
orig_km <- km
orig_summary <- summary(orig_km, times = time_points, extend = TRUE)
orig_se <- orig_summary$std.err
orig_surv <- orig_summary$surv

plot(time_points, orig_se, type = "l", col = "red", lwd = 2,
     xlab = "Time", ylab = "Standard Error", ylim = range(c(orig_se, bs_se)),
     main = "KM SE: Default vs. Bootstrap")
lines(time_points, bs_se, col = "blue", lwd = 2, lty = 2)
legend("topright", legend = c("Default KM SE", "Bootstrap SE"), col = c("red", "blue"), lty = c(1, 2))
```



Interpretation and comparison: - The red line shows the default standard errors estimated by the survfit function using the default Greenwood's formula - The blue line shows the bootstrapped standard errors with 200 iterations - The standard error increases as time progresses, which is expected in survival analysis because fewer individuals remain at risk at later time points (i.e., increased uncertainty). Hence it is constant at the end.

Bootstrap and Greenwood SE: - The two lines closely follow each other, showing high agreement between the analytical Greenwood method and the empirical bootstrap method. - This similarity validates the reliability of Greenwood's method. - Small deviations may occur due to sampling variability in the bootstrap.

Implication for Confidence Bands:

Since the SEs are quite similar between both methods, the ± 1 SE confidence bands computed using either method may be almost identical. This suggests the analytical method efficient in most clinical cases.

In conclusion, both standard error estimation techniques provide consistent results, but the bootstrap method gives a more robust SE estimate in regions with few patients or heavy censoring, which is common in clinical studies with long follow-up periods.

(c) Fit a Cox proportional hazards model that uses all of the predictors to predict survival. Summarize the main findings.

```
names(BrainCancer)
```

```
## [1] "sex"       "diagnosis" "loc"       "ki"        "gtv"       "stereo"
## [7] "status"    "time"
```

```
cox_model <- coxph(Surv(time, status) ~ ., data = BrainCancer)
summary(cox_model)
```

```
## Call:
## coxph(formula = Surv(time, status) ~ ., data = BrainCancer)
##
##   n= 87, number of events= 35
##    (1 observation deleted due to missingness)
##
##                       coef exp(coef) se(coef)      z Pr(>|z|)
## sexMale             0.18375   1.20171  0.36036  0.510  0.61012
## diagnosisLG glioma  0.91502   2.49683  0.63816  1.434  0.15161
## diagnosisHG glioma  2.15457   8.62414  0.45052  4.782 1.73e-06 ***
## diagnosisOther      0.88570   2.42467  0.65787  1.346  0.17821
## locSupratentorial   0.44119   1.55456  0.70367  0.627  0.53066
## ki                 -0.05496   0.94653  0.01831 -3.001  0.00269 **
## gtv                 0.03429   1.03489  0.02233  1.536  0.12466
## stereoSRT           0.17778   1.19456  0.60158  0.296  0.76760
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                    exp(coef) exp(-coef) lower .95 upper .95
## sexMale               1.2017     0.8321    0.5930    2.4352
## diagnosisLG glioma    2.4968     0.4005    0.7148    8.7215
## diagnosisHG glioma    8.6241     0.1160    3.5664   20.8546
## diagnosisOther        2.4247     0.4124    0.6678    8.8031
## locSupratentorial     1.5546     0.6433    0.3914    6.1741
## ki                    0.9465     1.0565    0.9132    0.9811
## gtv                   1.0349     0.9663    0.9906    1.0812
## stereoSRT             1.1946     0.8371    0.3674    3.8839
##
## Concordance= 0.794  (se = 0.04 )
## Likelihood ratio test= 41.37  on 8 df,   p=2e-06
## Wald test            = 38.7  on 8 df,   p=6e-06
## Score (logrank) test = 46.59  on 8 df,   p=2e-07
```

```
cox.zph(cox_model) #for proportional hazards assumption
```

```
##             chisq df    p
```

```
## sex       0.82855  1 0.36
## diagnosis 1.38692  3 0.71
## loc       0.65149  1 0.42
## ki        0.02801  1 0.87
## gtv       0.12701  1 0.72
## stereo    0.00623  1 0.94
## GLOBAL    3.61020  8 0.89
```

(d) Stratify the data by the value of ki. (Since only one observation has ki=40, you can group that observation together with the observations that have ki=60.) Plot Kaplan-Meier survival curves for each of the five strata, adjusted for the other predictors.

```
table(BrainCancer$ki)
```

```
##
##  40  60  70  80  90 100
##   1   5  14  36  27   5
```

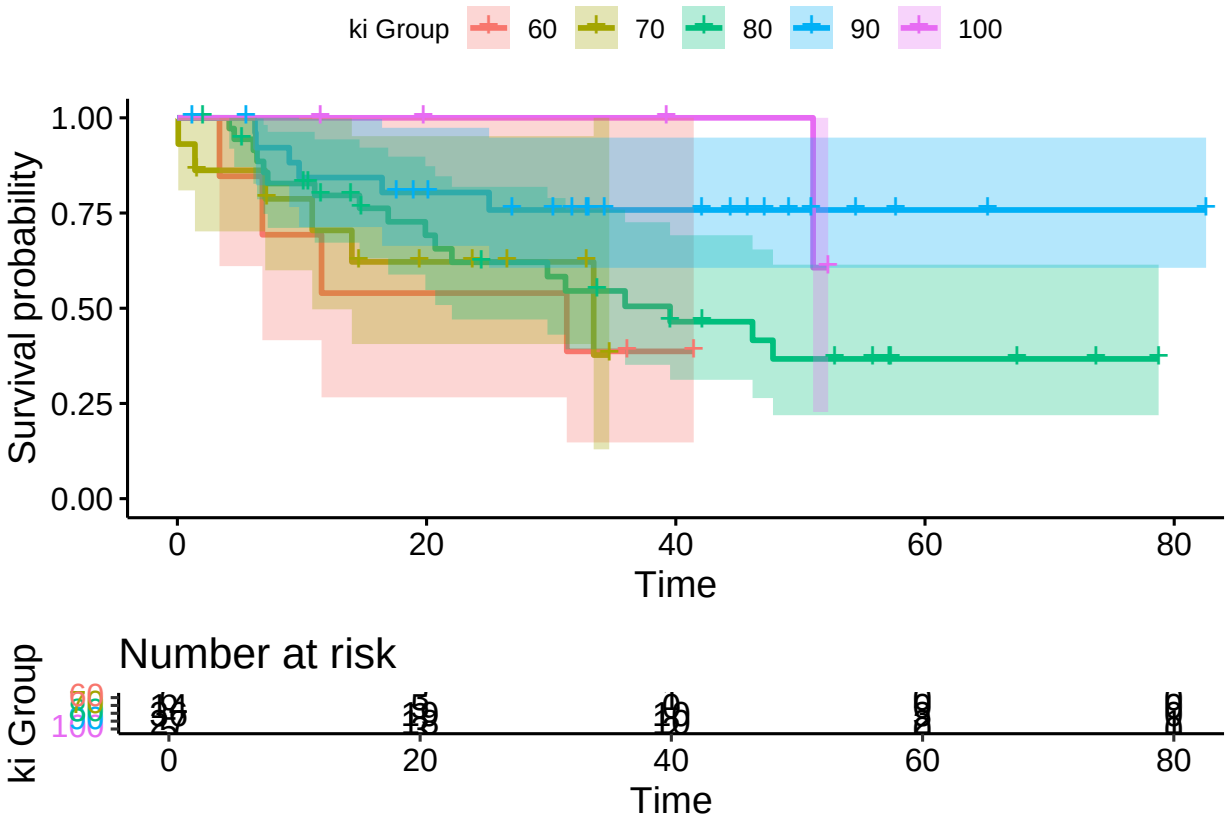
```
BrainCancer$ki_group <- ifelse(BrainCancer$ki == 40, 60, BrainCancer$ki)
BrainCancer$ki_group <- as.factor(BrainCancer$ki_group)
levels(BrainCancer$ki_group)
```

```
## [1] "60"  "70"  "80"  "90"  "100"
```

```
cox_strat <- coxph(Surv(time, status) ~ strata(ki_group), data = BrainCancer)
fit_strat <- survfit(cox_strat)
summary(cox_strat)
```

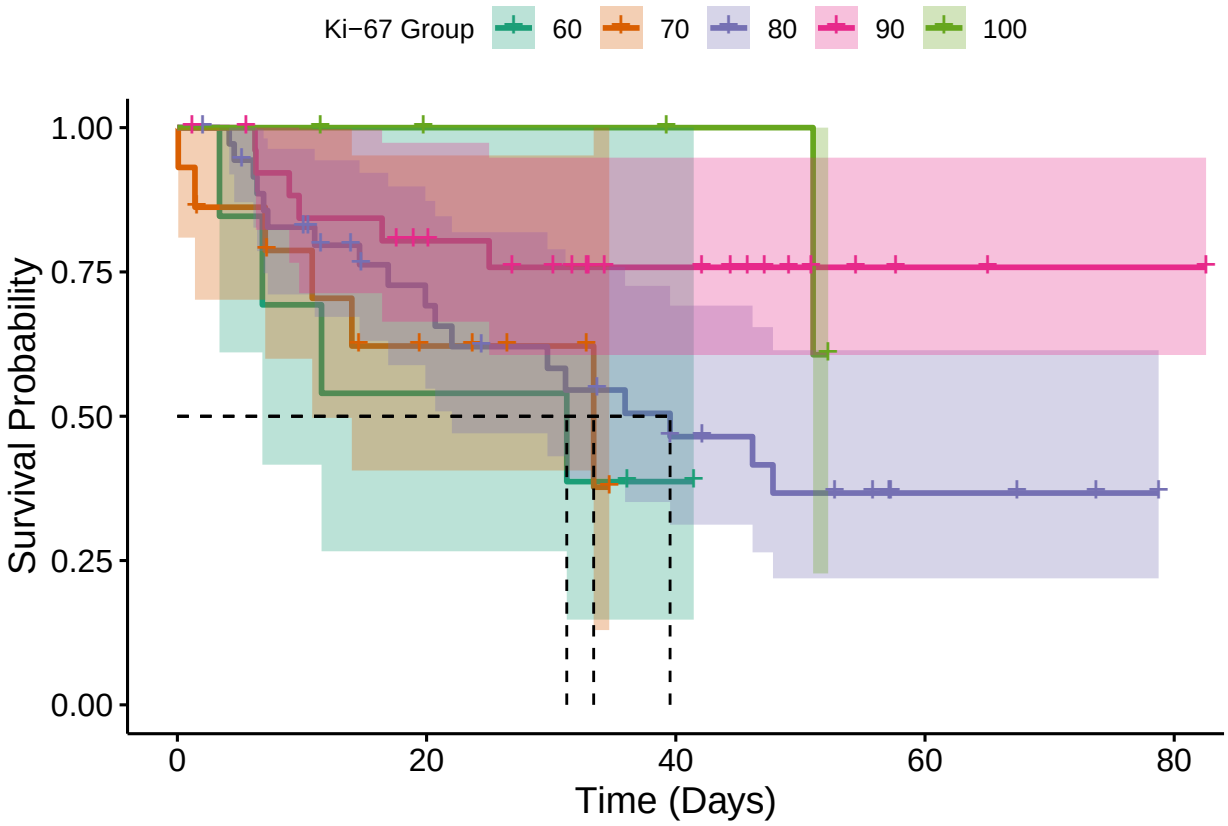
```
## Call:  coxph(formula = Surv(time, status) ~ strata(ki_group), data = BrainCancer)
##
## Null model
##   log likelihood= -91.47791
##   n= 88
```

```
ggsurvplot(fit_strat, data = BrainCancer,
           legend.title = "ki Group",
           conf.int = TRUE,
           risk.table = TRUE)
```




```
ggsurvplot(fit_strat, data = BrainCancer, conf.int = TRUE, xlab = "Time (Days)", ylab = "Survival Proba
```

```
## Warning in geom_segment(aes(x = 0, y = max(y2), xend = max(x1), yend = max(y2)), : All aesthetics hav
## i Please consider using `annotate()` or provide this layer with data containing
##   a single row.
## All aesthetics have length 1, but the data has 3 rows.
## i Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```


Ki−67 Group
60
70
80
90
100
1.00
0.75
0.50
0.25
0.00
Survival Probability
0
20
40
60
80
Time (Days)

Group Member – Payal Priyadarshini

Title- "A comprehensive benchmarking of machine learning algorithms and dimensionality reduction methods for drug sensitivity prediction"

https://doi.org/10.1093/bib/bbae242

Reference:

Lea Eckhart, Kerstin Lenhof, Lisa-Marie Rolli, Hans-Peter Lenhof, A comprehensive benchmarking of machine learning algorithms and dimensionality reduction methods for drug sensitivity prediction, *Briefings in Bioinformatics*, Volume 25, Issue 4, July 2024, bbae242, https://doi.org/10.1093/bib/bbae242

What is the main scientific question?

How can we best predict drug response in cancer cell lines using molecular biomarkers by benchmarking and optimizing different machine learning algorithms and dimension reduction techniques?

Write a summary (ca. 300 words):

This study benchmarks four commonly used machine learning algorithms—elastic nets, random forests, boosting trees, and neural networks—combined with nine dimensionality reduction strategies, across 179 anti-cancer drugs from the GDSC2 dataset. The dimensionality reduction methods include six feature selection (FS) approaches, such as minimum-redundancy-maximum-relevance (MRMR), correlation-based, and literature-based—and three feature extraction (FE) techniques like PCA, PASL, and autoencoders. Each method was evaluated across multiple input sizes (1–500 features). Over 16 million models were trained and assessed for prediction accuracy, interpretability, and computational efficiency. Elastic net consistently outperformed other ML models in both accuracy and runtime. Among DR methods, PCA and MRMR gave the best results, though PCA's interpretability is limited. Surprisingly, simple models using as few as 10–100 features often matched or exceeded the performance of complex deep learning models. Furthermore, standard models were more interpretable and faster to train. Additionally, the study showed that performance for sensitive cell lines typically underrepresented in datasets can be improved by changing hyperparameter tuning metrics. The results suggest that interpretable, efficient, and accurate drug response prediction is achievable with appropriate combinations of relatively simple models and DR strategies.

Keywords: Drug sensitivity prediction, machine learning, dimensionality reduction, gene expression, feature selection, feature extraction, elastic net, GDSC

What keywords did you use to find the paper?

-Machine learning drug response gene expression

-Omics data AND cancer drug response
-Drug sensitivity prediction machine learning

What reference paper did you use to find the paper?
It was easily available on academic.oup website and through:

1. Chiu et al. (2019) - Deep learning with multi-omics for drug response prediction
2. Li et al. (2023) - Evaluation of deep learning interpretability and performance

Data:

What type of data was generated?
Gene expression data and drug response data (log-transformed IC50 values)

What technologies were used to generate the data?
Microarray (Affymetrix Human Genome U219 Array)

How many samples were analysed?
Cell lines for 179 drugs, each with ≥600 samples

How many features (e.g. genes) were analysed?
Here features were genes- 17,419 genes

Data access: In what format is the data available (e.g. FASTQ)?
Gene expression data and drug response values are downloadable from the GDSC website (https://www.cancerrxgene.org/downloads/bulk_download) as processed matrices, in xls format.

Is any processed data available (e.g. mutation calls, count matrices,. . .)?
Yes — normalized gene expression, and IC50 log values for drug response

Are there any restrictions to data access?
No; open-access under Creative Commons License (CC BY 4.0)

Methodology:

What computational/statistical methods were used to analyze the data?

- Regression models: Elastic net, random forest, boosting trees, neural networks
- Dimensionality reduction models: Feature selection (e.g., correlation-based, MRMR, variance) and feature extraction (e.g., PCA, autoencoders, PASL)
- 5-fold cross-validation and test set evaluation
- Performance metrics: Mean squared error (MSE), interpretability, and runtime

Was a new method developed in the paper?

No entire novel ML model was developed so far, but they implemented an enrichment-based FS and evaluated MRMR via a heuristic and a quadratic programming approach.

What R/Python packages were used?

- **R:** glmnet, ranger, gbm, stats (PCA)
- **Python:** TensorFlow 1.13.1, Keras 2.3.1
- **Others:** GeneTrail (for GSEA)

Other:

How were the different data types integrated?

Primary analysis was based on gene expression. For one analysis, mutation data and multi-omics were used with a deep learning model

Was there anything special about the paper, e.g. single-cell analysis?

- Used bulk gene expression data from cell lines
- Comprehensive benchmarking across 16 million models
- Integration of interpretable Feature Selection approaches
- Emphasis on underrepresented sensitive cell lines

Can you imagine to do a similar analysis for your main project?

Yes, but I am planning to integrate cell line data for survival analysis and prognosis.

Related work: Compile a list of papers (2-3) that do a similar analysis.

1. Menden, M. P., Iorio, F., Garnett, M., McDermott, U., Benes, C. H., Ballester, P. J., & Saez-Rodriguez, J. (2013). Machine learning prediction of cancer cell sensitivity to drugs based on genomic and chemical properties. *PloS one*, *8*(4), e61318. https://doi.org/10.1371/journal.pone.0061318
2. Chiu, YC., Chen, HI.H., Zhang, T. *et al.* Predicting drug response of tumors from integrated genomic profiles by deep neural networks. *BMC Med Genomics* **12** (Suppl 1), 18 (2019). https://doi.org/10.1186/s12920-018-0460-9
3. Li, Y., Hostallero, D. E., & Emad, A. (2023). Interpretable deep learning architectures for improving drug response prediction performance: myth or reality?. *Bioinformatics (Oxford, England)*, *39*(6), btad390. https://doi.org/10.1093/bioinformatics/btad390

Are there any online databases where the type of omics data used in the paper is publicly available?

- GDSC (Genomics of Drug Sensitivity in Cancer): **https://www.cancerrxgene.org**
- CCLE (Cancer Cell Line Encyclopedia): **https://portals.broadinstitute.org/ccle**

Group Member – Renk Asik

Title:
Deep transfer learning of cancer drug responses by integrating bulk and single-cell RNA-seq data

Reference:
Chen, J., Wang, X., Ma, A. et al. Deep transfer learning of cancer drug responses by integrating bulk and single-cell RNA-seq data. *Nat Commun* 13, 6494 (2022). https://doi.org/10.1038/s41467-022-34277-7

Main scientific question:
Can deep transfer learning integrate bulk RNA-seq and single-cell RNA-seq data to accurately predict drug responses at the single-cell level in cancer?

Summary:

This study presents **scDEAL**, a deep transfer learning (DTL) framework that integrates bulk and single-cell RNA sequencing (scRNA-seq) data to predict cancer drug responses at the single-cell level. scDEAL addresses the challenge that traditional drug response prediction models—typically trained on bulk cell line data—are not directly applicable to single-cell data due to their complexity and heterogeneity.

The method utilizes bulk gene expression and drug response data from large-scale databases such as GDSC and CCLE to train a base model. Then, it employs denoising autoencoders (DAEs) and a domain-adaptive neural network (DaNN) to align low-dimensional gene expression features from bulk and single-cell data, enabling the transfer of learned drug-gene relationships.

The model was benchmarked on six publicly available scRNA-seq datasets treated with various drugs (*Cisplatin, Gefitinib, I-BET-762, Docetaxel, Erlotinib*), achieving high predictive accuracy (AUC = 0.898). Importantly, scDEAL also integrates cell clustering regularization to preserve single-cell heterogeneity during training and interprets predictions using integrated gradients to identify drug-resistance signature genes. Pseudotime analysis confirmed that predicted drug responses align with biological progression during drug treatment.

scDEAL demonstrates high interpretability and reliability across multiple datasets and conditions, even when bulk and single-cell data originate from different patients or species. The framework is expected to guide drug development, repurposing, and precision oncology by identifying key resistance mechanisms at the single-cell level.

Keywords:
drug response prediction, single-cell RNA-seq, bulk RNA-seq, cancer

Data:

> **Type of data generated:** Bulk and single-cell RNA-seq gene expression data, drug response annotations, integrated gradient scores for gene contribution.

Technologies used: RNA-seq (bulk and scRNA-seq), deep learning (DAE, MLP, DaNN), pseudotime inference (Monocle3)

Number of samples analyzed:

Bulk data: 1280 cell lines, 1557 drugs

Single-cell data: 6 scRNA-seq datasets (e.g., GSE110894 with 1419 cells)

Number of features analyzed: 15,962 genes (shared between datasets)

Data access:

Format:

Bulk expression: normalized expression matrices

scRNA-seq: count matrices

Processed data: binary drug response labels, integrated gradients, cluster assignments

Availability:

GDSC: https://www.cancerrxgene.org/

CCLE: https://depmap.org/portal/

GEO datasets: via accession numbers (e.g., GSE110894)

Restrictions: Publicly available, no access restrictions

Methodology:

Computational/statistical methods:

Deep Transfer Learning (DTL) using Domain-adaptive Neural Network (DaNN)

Denoising Autoencoders (DAEs) for low-dimensional feature extraction

Maximum Mean Discrepancy (MMD) loss to align feature distributions

Cross-entropy loss for label prediction

Integrated Gradients (IG) for model interpretability

Pseudotime inference using Monocle3

Novel method? Yes, scDEAL is a newly developed framework combining DTL and scRNA-seq data

R/Python packages used:

Python: imblearn, scikit-learn, scanpy, tensorflow/keras (implied)

Monocle3 (R) for pseudotime analysis

Other:

Integration approach: scDEAL aligns low-dimensional representations of bulk and single-cell gene expression via DTL to transfer drug-response knowledge.

Special features: Single-cell label-free prediction, integrated gradients for explainability, robust to cell-type heterogeneity, validated across multiple drugs and datasets.

Group Member – Leon Igal Inselberg Barbut

Non-oncology drugs are a source of previously unappreciated anti-cancer activity

The paper investigates whether non-oncology drugs can exhibit previously unrecognized anti-cancer activity and whether such effects can be predicted based on the genomic features of cancer cell lines.

The study presents a large-scale PRISM drug repurposing screen, testing 4,518 drugs (primarily non-oncology) against 578 human cancer cell lines to identify unexpected cytotoxic effects. The research leverages molecular barcoding to track drug-induced cell viability changes efficiently. Surprisingly, many non-oncology compounds selectively killes subsets of cancer cells, often in a manner predictable by genomic features. Several mechanisms were uncovered:

- PDE3A-SLFN12 protein interaction as a novel cytotoxic mechanism for PDE3A-expressing cancer cells.
- ABCB1 expression-dependent killing by the anti-inflammatory drug tepoxalin, reversing its expected role in drug resistance.
- Disulfiram sensitivity is linked to low expression of metallothionein genes, suggesting biomarker-based patient stratification.
- Vanadium compound cytotoxicity requiring SLC26A2 expression, pointing to an unknown sulfate transport dependency.

These discoveries highlight the potential of drug repurposing for cancer treatment, suggesting that systematic screening of existing pharmaceuticals can reveal novel oncology applications.

Keywords:

Drug repurposing, non-oncology drugs, cancer cell lines, genomic biomarkers, tepoxalin, drug response

Data:

- Type: Viability assays across cancer cell lines.
- Technologies: PRISM molecular barcoding, CRISPR-Cas9 genetic screens, LC-MS drug purity validation.
- Samples: 578 cancer cell lines spanning 24 tumor types.
- 4518 drugs from the Drug Repurposing Hub

Data access:
The data they use is publicly available in .csv format. Processed data is also available. There are normalized, filtered versions of the data also publicly available. There are no restrictions to access the data. Source: depmap.org

Methodology:

- Computational Methods: Random forest prediction (ATLANTIS algorithm), unsupervised clustering (UMAP).
- New Methods: No novel computational framework, but integration of PRISM barcoding with CRISPR screens.
- R/Python Packages: DeSeq2, ggplot2, dplyr, tidyr, scikit-learn, pandas, numpy, seaborn

Other:

- Data Integration: Combined PRISM drug response with genomic datasets.
- Special Features: Large-scale screening, potential clinical translation
- Application to Main Project: We could combine PRISM data with different types of omics data to predict if a drug could treat a specific cancer type (e.g. breast cancer)

Related Work:

- A single-cell analysis of breast cancer cell lines to study tumour heterogeneity and drug response

- Cancer Drug Response Profile scan (CDRscan): A Deep Learning Model That Predicts Drug Effectiveness from Cancer Genomic Signature